



Trading Bot System — Actual Trading Logic & Mechanics

Purpose

This document defines the **actual mechanics** of a Solana micro-cap trading bot: scoring, execution, risk management, and operational parameters. It is the system that the Convergence Framework is meant to debug and optimize.

Prerequisite: Read the "Trading Bot Convergence Framework — Corrected v2" before implementing this system. Every parameter here must be traced against the objective function $f(x)$ defined there.

1. MARKET UNIVERSE & DATA INGESTION

1.1 Token Universe

- **Chain:** Solana (SPL tokens)
- **Market cap filter:** \$50k – \$5M (micro-cap range)
- **Liquidity filter:** Minimum \$1,000 USD pool liquidity (Raydium, Orca, or Meteora pools)
- **Age filter:** Token launched $\geq$ 24 hours ago (avoid immediate rugs, allow price discovery)
- **Blacklist:** Exclude tokens with known scam patterns (mint authority still active, LP not burned, contract flagged)

1.2 Data Feeds

- **Price source:** Jupiter API (aggregated Solana DEX prices)

- **Liquidity source:** On-chain pool data (Raydium/Orca SDK)
- **Update frequency:** 5-second polling for active candidates; 30-second for screening
- **Historical data:** Minimum 4 hours of price history for scoring (descriptive inputs)
- **Forward signal data (if available):** Whale wallet tracking, social sentiment, exchange listing rumors — **must be validated separately before use**

2. SCORING SYSTEM (The "Score")

The score is a **descriptive composite** of past behavior. It has NO proven forward predictive validity unless separately validated (see OD-1 in Framework).

Components:

Component	Weight	Calculation	Rationale
Price momentum (1h)	25%	$(\text{price_now} / \text{price_1h_ago} - 1) * 100$ normalized to 0–100	Recent directional strength
Volume surge	20%	$\text{volume_1h} / \text{volume_4h_avg}$ normalized to 0–100	Attention/interest spike
Liquidity stability	15%	$1 - (\text{liquidity_std_4h} / \text{liquidity_mean_4h}) * 100$	Pool depth consistency
Holder growth	15%	$\text{new_holders_1h} / \text{total_holders} * 100$ (capped at 100)	Distribution widening
Social signal	15%	Mentions/minute normalized vs 4h baseline (if available)	External attention
Contract health	10%	Binary score (LP burned = 100, mint revoked = 100, verified = 100)	Safety floor

Calculation:

```
base_score = sum(component * weight for component, weight in components)
```

```
base_score = clamp(base_score, 0, 100)
```

2.2 Score Bonuses & Penalties

Bonuses:

- +10 if price momentum > +50% in 1h (strong breakout)
- +5 if volume surge > 5x average (viral attention)
- +5 if liquidity > \$50k (low slippage entry)

Penalties:

- -20 if liquidity < \$5k (high cost, likely un-winnable)
- -15 if contract health < 50 (safety risk)
- -10 if holder growth is negative (distribution contracting)
- -10 if price is at all-time high (limited upside, mean reversion risk)

Final score: `clamp(base_score + bonuses - penalties, 0, 100)`

3. ENTRY DECISION SYSTEM

3.1 Objective Function (from Framework)

```
f(x) = captureMult(score) * estimated_peak_pct(liq) - round_trip_cost(liq)
```

```
# captureMult(score) = 0.7 + 1.3 * clamp((score - 70) / 20, 0, 1.2)
```

```
# round_trip_cost(liq) = entry_slippage(liq) + exit_slippage(liq) + fees
```

3.2 Cost Model

Liquidity Tier	Entry Slippage	Exit Slippage	Fees	Total Cost
\$1k – \$5k	4.0%	4.0%	0.6%	8.6%
\$5k – \$10k	3.0%	3.0%	0.6%	6.6%
\$10k – \$25k	2.0%	2.0%	0.6%	4.6%
\$25k – \$100k	1.0%	1.0%	0.6%	2.6%
> \$100k	0.5%	0.5%	0.6%	1.6%

Note: These are empirical estimates. Update from real fill data after every 10 trades. If realized cost exceeds model by >20%, flag for model recalibration.

3.3 Estimated Peak (Descriptive Prior)

```
def estimateRealisticPeakPct(liq):
    """Returns a descriptive estimate of typical peak move for this liquidity tier.
    NOT a forecast. Used only in f(x) for feasibility screening."""
    if liq < 5000: return 3.0    # Low liq = limited upside, fast dump
    if liq < 10000: return 5.0  # Small cap, moderate pump
    if liq < 25000: return 8.0  # Mid-micro, room to run
    if liq < 50000: return 12.0 # Larger pool, more participants
    return 15.0                # High liq micro, rare but possible
```

Important: This is a historical prior based on observed distributions. It is NOT a prediction of what THIS token will do. It is used to answer: "Can this tier even theoretically produce enough move to beat costs?"

3.4 Entry Gates (Sequential)

A candidate must pass ALL gates to proceed to f(x) calculation:

Gate	Condition	Purpose
G1. Universe	Token in universe, not blacklisted, age >= 24h	Basic eligibility
G2. Quality Liquidity	Liquidity >= \$1,000	Avoid zero-liquidity traps
G3. Score Floor	Score >= 35	Minimum signal strength
G4. Sniper Override	(Score >= 85 AND Liquidity >= \$25k) OR (Score < 85)	High-score, high-liq gets sniper path
G5. Objective EV	$f(x) > +1.0\%$	The final decision: expected profit after costs

Entry path logic:

```

if not G1 or not G2 or not G3:
    return SKIP

if G4: # Sniper path
    # Sniper path allows faster entry but same f(x) gate
    pass

# Calculate f(x) – THIS IS THE DECISION
if f(x) > 1.0:
    return ENTER(size=position_size(liq, score))
else:
    return SKIP

```

Critical rule: Score alone NEVER triggers entry. Only $f(x) > 1.0\%$ triggers entry. Score is an input to $f(x)$, not the decision.

3.5 Position Sizing

```

def position_size(liq, score, conviction):
    """Conviction comes from f(x) margin, not score alone."""
    max_position = 0.05 * total_portfolio # 5% max per trade

```

```

    # Liquidity cap: never put more than 10% of pool liquidity
    y into one trade
    liquidity_cap = 0.10 * liq

    # Conviction scaling: higher f(x) margin = larger size (up
    p to max)
    conviction = f(x) / 0.10 # Normalize: 1.0% margin = 1.0x
    base, 5.0% = 5.0x
    base_size = 0.01 * total_portfolio # 1% base size
    conviction_size = base_size * clamp(conviction, 0.5, 5.0)

    return min(max_position, liquidity_cap, conviction_size)

```

4. EXIT SYSTEM

- ### 4.1 Stop Loss (-5%)
- **Hard stop:** Sell 100% of position if price falls -5% from entry price.
 - **Trigger type:** On-chain or off-chain monitoring with 5-second polling.
 - **Execution:** Market sell via Jupiter (best available route at time of trigger).
 - **Cost note:** The -5% stop is AFTER entry slippage. Actual realized loss from entry = -5% + entry_slippage. The stop price is computed as:

```

stop_price = entry_price * (1 - 0.05) # -5% from entry

```

4.2 Trailing Stop Arm (+8%)

- **Arm trigger:** When price reaches +8% from entry, activate trailing stop.
- **Trailing stop level:** 5% below the highest price reached since arm.
- **Example:**
 - Entry: \$1.00
 - Arm at: \$1.08 (+8%)
 - Price rises to \$1.20 → trailing stop moves to \$1.14 (5% below \$1.20)

- Price falls to \$1.14 → sell triggered
- **Purpose:** Capture larger moves while protecting profits.

4.3 Partial Take-Profit (+3%)

- **Trigger:** When price reaches +3% from entry.
- **Action:** Sell 50% of position.
- **Purpose:** Lock in some profit early, reduce exposure, improve risk-adjusted returns.
- **Important:** The +3% TP band must be OUTSIDE the round-trip cost. If cost at this liq is 4.6%, a +3% TP is **inside the cost spread** — this is un-winnable. The TP band must be > round_trip_cost. Adjust or skip partial TP if:

```
if 0.03 <= round_trip_cost(liq):
    partial_tp_enabled = False
```

4.4 Cooldown After Exit

- **After any exit (stop, TP, or trailing stop):** Token enters a 30-minute cooldown.
- **During cooldown:** Token is ineligible for re-entry, even if score spikes.
- **Purpose:** Prevent re-entry into the same volatility event, avoid whip-saw losses.
- **Exception:** Cooldown can be overridden if a new forward-predictive signal (validated separately) triggers with high confidence.

4.5 Exit Priority

When multiple exit conditions are triggered simultaneously:

1. **Stop loss** (highest priority — capital preservation)
2. **Partial take-profit** (if price hits +3% but not yet at trailing arm)
3. **Trailing stop** (if armed and breached)

If stop loss and partial TP trigger at the same time (e.g., price gap), stop loss wins.

5. RISK MANAGEMENT

5.1 Portfolio-Level Limits

- **Max drawdown:** Halt all trading if portfolio drawdown from daily high exceeds -10%.
- **Daily loss limit:** Stop trading for the day if realized losses exceed -5% of total portfolio.
- **Max open positions:** 10 simultaneous positions (to ensure adequate monitoring and liquidity).
- **Max exposure per token:** 5% of total portfolio (enforced by position sizing).

5.2 Liquidity Risk

- **Minimum pool liquidity:** \$1,000 (G2). Below this, slippage is too high and exit may be impossible.
- **Liquidity drop guard:** If pool liquidity drops by >50% while holding a position, trigger immediate exit. A rug pull or LP removal may be occurring.
- **No new entries during high volatility:** If Solana network TPS drops below 500 or block times exceed 2 seconds, pause new entries. Existing positions are monitored but new capital is not risked during degraded network conditions.

5.3 Smart Contract Risk

- **Contract health gate:** Score penalty of -15 if contract health < 50.
- **Mandatory checks before entry:**
 - Mint authority revoked (or supply fixed)
 - LP tokens burned (or locked)
 - Contract not flagged by community scanners (e.g., RugCheck, SolanaFM)
- **If any check fails:** Auto-blacklist token for 24 hours.

6. OPERATIONAL PARAMETERS

6.1 Monitoring & Alerting

- **Price polling:** 5-second intervals for active positions; 30-second for candidate screening.
- **Alert channels:** Telegram bot or Discord webhook for:
 - Entry executed
 - Exit triggered (stop, TP, trailing)
 - Risk limit breached (drawdown, daily loss)
 - Anomaly detected (liquidity drop, contract flag)
- **Log retention:** All trade logs, score snapshots, and $f(x)$ calculations retained for 90 days for post-hoc analysis.

6.2 Error Handling & Recovery

- **RPC failure:** Retry with exponential backoff (1s, 2s, 4s, 8s). After 4 failures, switch to backup RPC endpoint.
- **Jupiter swap failure:** Log error, retry once after 10 seconds. If still failing, mark position as "stuck" and alert for manual review.
- **Stale data guard:** If price data timestamp is >60 seconds old, flag as stale and pause new entries until fresh data arrives.
- **Crash recovery:** On restart, reload all open positions from on-chain state and resume monitoring. Do not re-enter positions that were already entered.

6.3 Rebalancing & Parameter Updates

- **Cost model recalibration:** After every 10 trades, compare realized costs to modeled costs. If error >20%, update slippage table.
- **Score weight review:** Monthly review of score component weights against realized win rates per component. If a component shows no correlation with success, reduce its weight or remove it.
- **Peak estimate review:** Quarterly review of `estimateRealisticPeakPct` against actual realized peak moves per liquidity tier. Adjust priors if systematically wrong.

- **Forward signal validation:** If a new predictive signal is introduced, require minimum 30 out-of-sample trades before it can be used in entry decisions. Log all signal-vs-outcome data.
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7. EXECUTION INTEGRATION (Solana)

7.1 Order Routing

- **Primary:** Jupiter Aggregator v6 (best price across Solana DEXs)
- **Slippage tolerance:** Set to `max(entry_slippage_model + 0.5%, 1.0%)`. Never exceed 2% to avoid bad fills.
- **Priority fee:** Use dynamic priority fee (0.00005–0.005 SOL) based on network congestion. If fee exceeds 0.01 SOL, pause entry and alert.

7.2 Wallet & Key Management

- **Trading wallet:** Dedicated hot wallet with only the funds allocated for active trading.
- **Cold wallet:** Majority of funds held in cold wallet; periodic transfers to hot wallet as needed.
- **Key storage:** Environment variable or secure key management service (e.g., AWS Secrets Manager, HashiCorp Vault). Never commit keys to code.

7.3 Transaction Confirmation

- **Minimum confirmations:** 1 (Solana finality is fast, but monitor for 5 seconds post-confirmation for any revert).
 - **Timeout:** If transaction is not confirmed within 30 seconds, consider it failed. Do not double-spend.
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8. TESTING & VALIDATION

8.1 Unit Tests (Required)

- **Score computation:** Test with known input data, verify score matches expected output.

- **f(x) calculation:** Test with all liquidity tiers, verify sign of $f(x)$ matches expected profitability.
- **Position sizing:** Test with extreme values (liq=\$1k, liq=\$1M, score=0, score=100).
- **Gate logic:** Test every gate individually with passing and failing inputs.
- **Exit triggers:** Test stop, trailing stop, and partial TP with simulated price paths.

8.2 Backtest (Before Paper Trading)

- **Historical data:** Minimum 30 days of Solana micro-cap data.
- **Simulation:** Replay all candidate tokens through the full system (scoring, gates, entry, exit).
- **Output:** Generate P&L, win rate, average return, max drawdown, cost accuracy.
- **Acceptance:** Backtest must show positive mean return per trade (before live costs) to justify paper trading.

8.3 Paper Trading (Before Live)

- **Environment:** Real market data, simulated execution (no real capital).
- **Duration:** Minimum 30 trades or 14 days, whichever is longer.
- **Metrics to track:**
 - Realized $f(x)$ vs expected $f(x)$ (model accuracy)
 - Win rate and average win/loss size
 - Cost model accuracy (realized vs modeled slippage)
 - Score correlation with outcome (is score actually predictive?)
- **Decision:** Proceed to live trading ONLY if:
 - Realized $\text{mean}(f(x)) > +1.0\%$
 - Cost model error $< 20\%$
 - Max drawdown $< 10\%$

- No critical bugs found during paper period

8.4 Live Trading — Gradual Ramp

- **Phase 1:** 10% of target position size for first 10 trades.
- **Phase 2:** 50% of target position size for next 20 trades.
- **Phase 3:** 100% of target position size after 30 successful trades with positive realized $f(x)$.
- **Kill switch:** If live performance deviates from paper performance by >30% (worse), revert to paper and debug.

9. KNOWN ISSUES & WATCH LIST

ID	Issue	Status	Mitigation
OD-1	No validated forward-predictive input	Open	Score is descriptive only. Do not trade unless forward signal is validated or descriptive inputs show empirical predictive validity.
OD-2	Low-liquidity un-winnable entries	Fixed	$f(x)$ gate + liquidity floor prevents entries where cost > expected peak.
OD-3	Partial-TP at +3% may be inside cost spread	Watch	Disable partial TP if <code>round_trip_cost >= 3%</code> . Review per-trade.
OD-4	Score not EV-validated	Watch	Score must inform $f(x)$, never be the decision. Review correlation monthly.
OD-5	Conviction double-counting (original framework)	Fixed	$f(x)$ uses captureMult only. Conviction is not a separate factor.
OD-6	Solana network congestion	Ongoing	Pause entries if TPS < 500 or block times > 2s. Monitor Jupiter API health.

ID	Issue	Status	Mitigation
OD-7	Rug pull / contract exploit	Ongoing	Contract health gate + blacklist + LP burn check. Liquidity drop guard.

10. SUMMARY

- Ingest:** Filter `universe` (Solana, \$50k-\$5M MC, \$1k+ liq, 24h+ age)
- Score:** Compute descriptive `score` (0-100) from momentum, volume, liquidity, holders, social, contract
- Gate:** Pass G1-G4 (universe, liquidity, score floor, sniper override)
- Decide:** Calculate $f(x) = \text{captureMult}(\text{score}) * \text{estimated_peak}(\text{liq}) - \text{round_trip_cost}(\text{liq})$
- Enter:** If $f(x) > 1.0\%$, enter with `position_size`(liq, score, conviction)
- Monitor:** 5-second polling, track price vs stop, trailing arm, partial TP
- Exit:** On `stop` (-5%), partial `TP` (+3%, 50%), or trailing `stop` (armed at +8%, 5% trail)
- Cooldown:** 30-minute post-exit cooldown
- Risk:** Halt if drawdown >10%, daily loss >5%, or liq drops >50%
- Validate:** Paper trade 30+ trades, then live ramp 10% → 50% → 100%

This document is the system. The Convergence Framework is the debugger. Both must be maintained together.